



Tree-Partitions with Bounded Degree Trees

Marc Distel and David R. Wood

Abstract A *tree-partition* of a graph G is a partition of $V(G)$ such that identifying the vertices in each part gives a tree. It is known that every graph with treewidth k and maximum degree Δ has a tree-partition with parts of size $O(k\Delta)$. We prove the same result with the extra property that the underlying tree has maximum degree $O(\Delta)$.

1 Introduction

For a graph G and a tree T , a *T -partition* of G is a partition $(V_x : x \in V(T))$ of $V(G)$ indexed by the nodes of T , such that for every edge vw of G , if $v \in V_x$ and $w \in V_y$, then $x = y$ or $xy \in E(T)$. The *width* of a T -partition is $\max\{|V_x| : x \in V(T)\}$. The *tree-partition-width*¹ of a graph G is the minimum width of a tree-partition of G .

Tree-partitions were independently introduced by Seese [31] and Halin [24], and have since been widely investigated [6–8, 15, 16, 21, 32, 33]. Applications of tree-partitions include graph drawing [12, 14, 19, 20, 34], graphs of linear growth [11], nonrepetitive graph colouring [2], clustered graph colouring [1, 27], monadic second-order logic [26], network emulations [3, 4, 9, 22], statistical learning theory [35], and the edge-Erdős-Pósa property [13, 23, 28]. Tree-partitions are also related to graph product structure theory since a graph G has a T -partition of width at most k if and only if G is isomorphic to a subgraph of $T \boxtimes K_k$ for some tree T ; see [10, 17, 18] for example.

Marc Distel, David R. Wood

School of Mathematics, Monash University, Melbourne, Australia

e-mail: {marc.distel, david.wood}@monash.edu

¹ Tree-partition-width has also been called *strong treewidth* [7, 31].

Bounded tree-partition-width implies bounded treewidth², as noted by Seese [31]. In particular, for every graph G ,

$$\text{tw}(G) \leq 2\text{tpw}(G) - 1.$$

Of course, $\text{tw}(T) = \text{tpw}(T) = 1$ for every tree T . But in general, $\text{tpw}(G)$ can be much larger than $\text{tw}(G)$. For example, fan graphs on n vertices have treewidth 2 and tree-partition-width $\Omega(\sqrt{n})$. On the other hand, the referee of [15] showed that if the maximum degree and treewidth are both bounded, then so is the tree-partition-width, which is one of the most useful results about tree-partitions. A graph G is *trivial* if $E(G) = \emptyset$. Let $\Delta(G)$ be the maximum degree of G .

Theorem 1 ([15]). *For any non-trivial graph G ,*

$$\text{tpw}(G) \leq 24(\text{tw}(G) + 1)\Delta(G).$$

Theorem 1 is stated in [15] with “ $\text{tw}(G)$ ” instead of “ $\text{tw}(G) + 1$ ”, but a close inspection of the proof shows that “ $\text{tw}(G) + 1$ ” is needed. Wood [33] showed that Theorem 1 is best possible up to the multiplicative constant, and also improved the constant 24 to $9 + 6\sqrt{2} \approx 17.48$.

This paper considers the maximum degree of T in a T -partition. Consider a tree-partition $(B_x : x \in V(T))$ of a graph G with width k . For each node $x \in V(T)$, there are at most $\sum_{v \in B_x} \deg(v)$ edges between B_x and $G - B_x$. Thus we may assume that $\deg_T(x) \leq |B_x|\Delta(G) \leq k\Delta(G)$, otherwise delete an ‘unused’ edge of T and add an edge to T between leaf vertices of the resulting component subtrees. It follows that if $\text{tpw}(G) \leq k$ then G has a T -partition of width at most k for some tree T with maximum degree at most $\max\{k\Delta(G), 2\}$. By Theorem 1, every graph G has a T -partition of width at most $24(\text{tw}(G) + 1)\Delta(G)$ for some tree T with maximum degree at most $24(\text{tw}(G) + 1)\Delta(G)^2$. This fact has been used in several applications of Theorem 1 (see [12, 20] for example). The following theorem improves this upper bound on $\Delta(T)$. Indeed, $\Delta(T)$ is independent of $\text{tw}(G)$.

Theorem 2. *Every non-trivial graph G has a T -partition of width at most*

$$18(\text{tw}(G) + 1)\Delta(G)$$

for some tree T with $\Delta(T) \leq 6\Delta(G)$.

Theorem 2 enables a $\text{tw}(G)\Delta(G)^2$ term to be replaced by a $\Delta(G)$ term in various results [12, 20].

² A *tree-decomposition* of a graph G is a collection $(B_x \subseteq V(G) : x \in V(T))$ of subsets of $V(G)$ (called *bags*) indexed by the nodes of a tree T , such that: (a) for every edge $uv \in E(G)$, some bag B_x contains both u and v ; and (b) for every vertex $v \in V(G)$, the set $\{x \in V(T) : v \in B_x\}$ induces a non-empty subtree of T . The *width* of a tree-decomposition is the size of the largest bag, minus 1. The *treewidth* of a graph G , denoted by $\text{tw}(G)$, is the minimum width of a tree-decomposition of G . Treewidth is the standard measure of how similar a graph is to a tree. Indeed, a connected graph has treewidth 1 if and only if it is a tree. Treewidth is of fundamental importance in structural and algorithmic graph theory; see [25, 29?] for surveys.

As mentioned above, Wood [33] improved the constant 24 to $9 + 6\sqrt{2}$ in Theorem 1. By tweaking the constants in the proof of Theorem 2, we match this constant with a small increase in the bound on $\Delta(T)$; see Section 3. We choose to first present the proof with integer coefficients for ease of understanding.

Our final result shows that the linear upper bound on $\Delta(T)$ in Theorem 2 is best possible even for trees.

Proposition 3. *For any integer $\Delta \geq 3$ there exist $\alpha > 0$ such that there are infinitely many trees X with maximum degree Δ such that for every tree T with maximum degree less than Δ , every T -partition of X has width at least $|V(X)|^\alpha$. Moreover, if $\Delta = 3$ then α can be taken to be arbitrarily close to 1.*

2 Main Proofs

Theorem 2 is implied by the following lemma. The proof is identical to the proof of Theorem 1, except that we pay attention to $\Delta(T)$. Let $\mathbb{N} = \{1, 2, \dots\}$.

Lemma 4. *Fix $k, d \in \mathbb{N}$. Let G be a graph with $\text{tw}(G) \leq k - 1$ and $\Delta(G) \leq d$. Then G has a tree-partition $(B_x : x \in V(T))$ of width at most $18kd$ such that $\Delta(T) \leq 6d$. Moreover, for any set $S \subseteq V(G)$ with $4k \leq |S| \leq 12kd$, there exists a tree-partition $(B_x : x \in V(T))$ of G with width at most $18kd$, such that $\Delta(T) \leq 6d$ and there exists $z \in V(T)$ such that:*

- $S \subseteq B_z$,
- $|B_z| \leq \frac{3}{2}|S| - 2k$,
- $\deg_T(z) \leq \frac{|S|}{2k} - 1$.

Proof. We proceed by induction on $|V(G)|$.

Case 1. $|V(G)| < 4k$: Then S is not specified. Let T be the 1-vertex tree with $V(T) = \{x\}$, and let $B_x := V(G)$. Then $(B_x : x \in V(T))$ is the desired tree-partition, since $|B_x| = |V(G)| < 4k \leq 18kd$ and $\Delta(T) = 0 \leq 6d$.

Now assume that $|V(G)| \geq 4k$. If S is not specified, then let S be any set of $4k$ vertices in G .

Case 2. $|V(G - S)| \leq 18kd$: Let T be the 2-vertex tree with $V(T) = \{y, z\}$ and $E(T) = \{yz\}$. Note that $\Delta(T) = 1 \leq 6d$ and $\deg_T(z) = 1 \leq \frac{|S|}{2k} - 1$. Let $B_z := S$ and $B_y := V(G - S)$. Thus $|B_z| = |S| \leq \frac{3}{2}|S| - 2k \leq 18kd$ and $|B_y| \leq |V(G - S)| \leq 18kd$. Hence $(B_x : x \in V(T))$ is the desired tree-partition of G .

Now assume that $|V(G - S)| \geq 18kd$.

Case 3. $4k \leq |S| \leq 12kd$: Let $S' := \bigcup \{N_G(v) \setminus S : v \in S\}$. Thus $|S'| \leq d|S| \leq 12kd$. If $|S'| < 4k$ then add $4k - |S'|$ vertices from $V(G - S - S')$ to S' , so that $|S'| = 4k$. This is well-defined since $|V(G - S)| \geq 18kd \geq 4k$, implying $|V(G - S - S')| \geq 4k - |S'|$. By induction, there exists a tree-partition $(B_x : x \in V(T'))$ of $G - S$ with width at most $18kd$, such that $\Delta(T') \leq 6d$ and there exists $z' \in V(T')$ such that:

- $S' \subseteq B_{z'}$,

- $|B_{z'}| \leq \frac{3}{2}|S'| - 2k \leq 18kd - 2k$,
- $\deg_{T'}(z') \leq \frac{|S'|}{2k} - 1 \leq 6d - 1$.

Let T be the tree obtained from T' by adding one new node z adjacent to z' . Let $B_z := S$. So $(B_x : x \in V(T))$ is a tree-partition of G with width at most $\max\{18kd, |S|\} \leq \max\{18kd, 12k\} = 18kd$. By construction, $\deg_T(z) = 1 \leq \frac{|S|}{2k} - 1$ and $\deg_T(z') = \deg_{T'}(z') + 1 \leq (6d - 1) + 1 = 6d$. Every other vertex in T has the same degree as in T' . Hence $\Delta(T) \leq 6d$, as desired. Finally, $S = B_z$ and $|B_z| = |S| \leq \frac{3}{2}|S| - 2k$.

Case 4. $12k \leq |S| \leq 12kd$: By the separator lemma of Robertson and Seymour [30, (2.6)], there are induced subgraphs G_1 and G_2 of G with $G_1 \cup G_2 = G$ and $|V(G_1 \cap G_2)| \leq k$, where $|S \cap V(G_i)| \leq \frac{2}{3}|S|$ for each $i \in \{1, 2\}$. Let $S_i := (S \cap V(G_i)) \cup V(G_1 \cap G_2)$ for each $i \in \{1, 2\}$.

We now bound $|S_i|$. For a lower bound, since $|S \cap V(G_1)| \leq \frac{2}{3}|S|$, we have $|S_2| \geq |S \setminus V(G_1)| \geq \frac{1}{3}|S| \geq 4k$. By symmetry, $|S_1| \geq 4k$. For an upper bound, $|S_i| \leq \frac{2}{3}|S| + k \leq 8kd + k \leq 12kd$. Also note that $|S_1| + |S_2| \leq |S| + 2k$.

We have shown that $4k \leq |S_i| \leq 12kd$ for each $i \in \{1, 2\}$. Thus we may apply induction to G_i with S_i the specified set. Hence there exists a tree-partition $(B'_x : x \in V(T_i))$ of G_i with width at most $18kd$, such that $\Delta(T_i) \leq 6d$ and there exists $z_i \in V(T_i)$ such that:

- $S_i \subseteq B_{z_i}$,
- $|B_{z_i}| \leq \frac{3}{2}|S_i| - 2k$,
- $\deg_{T_i}(z_i) \leq \frac{|S_i|}{2k} - 1$.

Let T be the tree obtained from the disjoint union of T_1 and T_2 by merging z_1 and z_2 into a vertex z . Let $B_z := B_{z_1}^1 \cup B_{z_2}^2$. Let $B_x := B'_x$ for each $x \in V(T_i) \setminus \{z_i\}$. Since $G = G_1 \cup G_2$ and $V(G_1 \cap G_2) \subseteq B_{z_1}^1 \cap B_{z_2}^2 \subseteq B_z$, we have that $(B_x : x \in V(T))$ is a tree-partition of G . By construction, $S \subseteq B_z$ and since $V(G_1 \cap G_2) \subseteq B_{z_i}^i$ for each i ,

$$\begin{aligned}
 |B_z| &\leq |B_{z_1}^1| + |B_{z_2}^2| - |V(G_1 \cap G_2)| \\
 &\leq \left(\frac{3}{2}|S_1| - 2k\right) + \left(\frac{3}{2}|S_2| - 2k\right) - |V(G_1 \cap G_2)| \\
 &= \frac{3}{2}(|S_1| + |S_2|) - 4k - |V(G_1 \cap G_2)| \\
 &\leq \frac{3}{2}(|S| + 2|V(G_1 \cap G_2)|) - 4k - |V(G_1 \cap G_2)| \\
 &\leq \frac{3}{2}|S| + 2|V(G_1 \cap G_2)| - 4k \\
 &\leq \frac{3}{2}|S| - 2k \\
 &< 18kd.
 \end{aligned}$$

Every other part has the same size as in the tree-partition of G_1 or G_2 . So this tree-partition of G has width at most $18kd$. Note that

$$\begin{aligned}
 \deg_T(z) &= \deg_{T_1}(z_1) + \deg_{T_2}(z_2) \leq \left(\frac{|S_1|}{2k} - 1\right) + \left(\frac{|S_2|}{2k} - 1\right) \\
 &= \frac{|S_1| + |S_2|}{2k} - 2 \\
 &\leq \frac{|S| + 2k}{2k} - 2
 \end{aligned}$$

$$\begin{aligned}
&= \frac{|S|}{2k} - 1 \\
&< 6d.
\end{aligned}$$

Every other node of T has the same degree as in T_1 or T_2 . Thus $\Delta(T) \leq 6d$. This completes the proof. $\square$

We now prove the lower bound. For $\Delta, d \in \mathbb{N}$ with $\Delta \geq 2$, let $X_{\Delta,d}$ be the tree rooted at a vertex r such that every leaf is at distance d from r and every non-leaf vertex has degree Δ . Observe that $X_{\Delta,d}$ has the maximum number of vertices in a tree with maximum degree Δ and radius d , where

$$|V(X_{\Delta,d})| = 1 + \Delta \sum_{i=0}^{d-1} (\Delta - 1)^i.$$

Note that $|V(X_{2,d})| = 2d + 1$, and if $\Delta \geq 3$ then

$$(\Delta - 1)^d \leq |V(X_{\Delta,d})| = 1 + \frac{\Delta}{\Delta - 2} ((\Delta - 1)^d - 1) \leq 3(\Delta - 1)^d.$$

Proposition 3. *For any integer $\Delta \geq 3$ there exist $\alpha > 0$ such that there are infinitely many trees X with maximum degree Δ such that for every tree T with maximum degree less than Δ , every T -partition of X has width at least $|V(X)|^\alpha$. Moreover, if $\Delta = 3$ then α can be taken to be arbitrarily close to 1.*

Proof. First suppose that $\Delta \geq 4$. Let $d_0 \in \mathbb{N}$ be sufficiently large so that $(\frac{\Delta-1}{\Delta-2})^{d_0} > 3$. Let $\alpha := 1 - \log_{\Delta-1}(3^{1/d_0}(\Delta-2))$, which is positive by the choice of d_0 . Let $d \in \mathbb{N}$ with $d \geq d_0$. It follows that $(\Delta-1)^{(1-\alpha)d} \geq 3(\Delta-2)^d$. Consider any tree-partition $(B_u : u \in V(T))$ of $X_{\Delta,d}$, where T is any tree with maximum degree at most $\Delta-1$. Let z be the vertex of T such that the root $r \in B_z$. Since adjacent vertices in $X_{\Delta,d}$ belong to adjacent parts or the same part in T , every vertex in T is at distance at most d from z . Thus T has radius at most d , and

$$|V(T)| \leq |V(X_{\Delta-1,d})| \leq 3(\Delta-2)^d \leq (\Delta-1)^{(1-\alpha)d} \leq |V(X_{\Delta,d})|^{1-\alpha}.$$

By the pigeon-hole principle, there is a vertex $u \in V(T)$ such that $|B_u| \geq \frac{|V(X_{\Delta,d})|}{|V(T)|} \geq |V(X_{\Delta,d})|^\alpha$.

Now assume that $\Delta = 3$. Let $\alpha \in (0, 1)$, let $d_0 \in \mathbb{N}$ be sufficiently large so that $2d_0 + 1 \leq 2^{(1-\alpha)d_0}$, and let $d \geq d_0$. So $2d + 1 \leq 2^{(1-\alpha)d}$. Consider any tree-partition $(B_u : u \in V(T))$ of $X_{3,d}$, where T is any tree with maximum degree at most 2. By the argument above, T has radius at most d , implying

$$|V(T)| \leq |V(X_{2,d})| = 2d + 1 \leq 2^{(1-\alpha)d} \leq |V(X_{3,d})|^{1-\alpha}.$$

Again, there is a vertex $u \in V(T)$ such that $|B_u| \geq \frac{|V(X_{3,d})|}{|V(T)|} \geq |V(X_{3,d})|^\alpha$. $\square$

3 Tweaking the Constants

Consider the following generalisation of Lemma 4.

Lemma 5. Fix $k, d \in \mathbb{N}$ and $\alpha \in \mathbb{R}$ with $\alpha > 2$. Let G be a graph with $\text{tw}(G) \leq k - 1$ and $\Delta(G) \leq d$. Then G has a tree-partition $(B_x : x \in V(T))$ of width at most

$$\frac{3\alpha(\alpha-1)}{\alpha-2}kd - \frac{\alpha}{\alpha-2}k$$

such that $\Delta(T) \leq \frac{3\alpha}{\alpha-2}d + \frac{\alpha-4}{\alpha-2}$. Moreover, for any set $S \subseteq V(G)$ with $\alpha k \leq |S| \leq 3\alpha kd$, there exists a tree-partition $(B_x : x \in V(T))$ of G with width at most $\frac{3\alpha(\alpha-1)}{\alpha-2}kd - \frac{\alpha}{\alpha-2}k$, such that $\Delta(T) \leq \frac{3\alpha}{\alpha-2}d + \frac{\alpha-4}{\alpha-2}$ and there exists $z \in V(T)$ such that:

- $S \subseteq B_z$,
- $|B_z| \leq \frac{\alpha-1}{\alpha-2}|S| - \frac{\alpha}{\alpha-2}k$,
- $\deg_T(z) \leq \frac{1}{(\alpha-2)k}|S| - \frac{2}{\alpha-2}$.

Proof. We proceed by induction on $|V(G)|$.

Case 1. $|V(G)| < \alpha k$: Then S is not specified. Let T be the 1-vertex tree with $V(T) = \{x\}$, and let $B_x := V(G)$. Then $(B_x : x \in V(T))$ is the desired tree-partition, since $|B_x| = |V(G)| < \alpha k \leq \frac{3\alpha(\alpha-1)}{\alpha-2}kd - \frac{\alpha}{\alpha-2}k$ and $\Delta(T) = 0 \leq \frac{3\alpha}{\alpha-2}d + \frac{\alpha-4}{\alpha-2}$.

Now assume that $|V(G)| \geq \alpha k$. If S is not specified, then let S be any set of $\lceil \alpha k \rceil$ vertices in G (implying $|S| \leq 3\alpha kd$).

Case 2. $|V(G-S)| \leq \frac{3\alpha(\alpha-1)}{\alpha-2}kd - \frac{\alpha}{\alpha-2}k$: Let T be the 2-vertex tree with $V(T) = \{y, z\}$ and $E(T) = \{yz\}$. Note that $\Delta(T) = 1 \leq \frac{3\alpha}{\alpha-2}d + \frac{\alpha-4}{\alpha-2}$ and $\deg_T(z) = 1 \leq \frac{1}{(\alpha-2)k}|S| - \frac{2}{\alpha-2}$. Let $B_z := S$ and $B_y := V(G-S)$. Thus $|B_z| = |S| \leq \frac{\alpha-1}{\alpha-2}|S| - \frac{\alpha}{\alpha-2}k \leq \frac{3\alpha(\alpha-1)}{\alpha-2}kd - \frac{\alpha}{\alpha-2}k$ and $|B_y| \leq |V(G-S)| \leq \frac{3\alpha(\alpha-1)}{\alpha-2}kd$. Hence $(B_x : x \in V(T))$ is the desired tree-partition of G .

Now assume that $|V(G-S)| \geq \frac{3\alpha(\alpha-1)}{\alpha-2}kd - \frac{\alpha}{\alpha-2}k$.

Case 3. $\alpha k \leq |S| \leq 3\alpha k$: Let $S' := \bigcup_{v \in S} N_G(v) \setminus S$. Thus $|S'| \leq d|S| \leq 3\alpha kd$. If $|S'| < \alpha k$ then add $\alpha k - |S'|$ vertices from $V(G-S-S')$ to S' , so that $|S'| = \alpha k$. This is well-defined since $|V(G-S)| \geq \frac{3\alpha(\alpha-1)}{\alpha-2}kd - \frac{\alpha}{\alpha-2}k \geq \alpha k$, implying $|V(G-S-S')| \geq \alpha k - |S'|$. By induction, there exists a tree-partition $(B_x : x \in V(T'))$ of $G-S$ with width at most $\frac{3\alpha(\alpha-1)}{\alpha-2}kd - \frac{\alpha}{\alpha-2}k$, such that $\Delta(T') \leq \frac{3\alpha}{\alpha-2}d + \frac{\alpha-4}{\alpha-2}$ and there exists $z' \in V(T')$ such that:

- $S' \subseteq B_{z'}$,
- $|B_{z'}| \leq \frac{\alpha-1}{\alpha-2}|S'| - \frac{\alpha}{\alpha-2}k \leq \frac{3\alpha(\alpha-1)}{\alpha-2}kd - \frac{\alpha}{\alpha-2}k$,
- $\deg_{T'}(z') \leq \frac{1}{(\alpha-2)k}|S'| - \frac{2}{\alpha-2} \leq \frac{3\alpha}{\alpha-2}d - \frac{2}{\alpha-2}$.

Let T be the tree obtained from T' by adding one new node z adjacent to z' . Let $B_z := S$. So $(B_x : x \in V(T))$ is a tree-partition of G with width at most

$$\max\left\{\frac{3\alpha(\alpha-1)}{\alpha-2}kd - \frac{\alpha}{\alpha-2}k, |S|\right\} \leq \max\left\{\frac{3\alpha(\alpha-1)}{\alpha-2}kd - \frac{\alpha}{\alpha-2}k, 3\alpha k\right\} = \frac{3\alpha(\alpha-1)}{\alpha-2}kd - \frac{\alpha}{\alpha-2}k.$$

By construction,

$$\begin{aligned}\deg_T(z) &= 1 \leq \frac{1}{(\alpha-2)k}|S| - \frac{2}{\alpha-2} \quad \text{and} \\ \deg_T(z') &= \deg_{T'}(z') + 1 \leq \frac{3\alpha}{\alpha-2}d - \frac{2}{\alpha-2} + 1 = \frac{3\alpha}{\alpha-2}d + \frac{\alpha-4}{\alpha-2}.\end{aligned}$$

Every other vertex in T has the same degree as in T' . Hence $\Delta(T) \leq \frac{3\alpha}{\alpha-2}d + \frac{\alpha-4}{\alpha-2}$, as desired. Finally, $S = B_z$ and $|B_z| = |S| \leq \frac{\alpha-1}{\alpha-2}|S| - \frac{\alpha}{\alpha-2}k$.

Case 4. $3\alpha k \leq |S| \leq 3\alpha kd$: By the separator lemma of Robertson and Seymour [30, (2.6)], there are induced subgraphs G_1 and G_2 of G with $G_1 \cup G_2 = G$ and $|V(G_1 \cap G_2)| \leq k$, where $|S \cap V(G_i)| \leq \frac{2}{3}|S|$ for each $i \in \{1, 2\}$. Let $S_i := (S \cap V(G_i)) \cup V(G_1 \cap G_2)$ for each $i \in \{1, 2\}$.

We now bound $|S_i|$. For a lower bound, since $|S \cap V(G_1)| \leq \frac{2}{3}|S|$, we have $|S_2| \geq |S \setminus V(G_1)| \geq \frac{1}{3}|S| \geq \frac{1}{3}3\alpha k \geq \alpha k$. By symmetry, $|S_1| \geq \alpha k$. For an upper bound, $|S_i| \leq \frac{2}{3}|S| + k \leq 2\alpha kd + k \leq 3\alpha kd$. Also note that $|S_1| + |S_2| \leq |S| + 2|V(G_1 \cap G_2)| \leq |S| + 2k$.

We have shown that $\alpha k \leq |S_i| \leq 3\alpha kd$ for each $i \in \{1, 2\}$. Thus we may apply induction to G_i with S_i the specified set. Hence there exists a tree-partition $(B_x^i : x \in V(T_i))$ of G_i with width at most $\frac{3\alpha(\alpha-1)}{\alpha-2}kd - \frac{\alpha}{\alpha-2}k$, such that $\Delta(T_i) \leq \frac{3\alpha}{\alpha-2}d + \frac{\alpha-4}{\alpha-2}$ and there exists $z_i \in V(T_i)$ such that:

- $S_i \subseteq B_{z_i}^i$,
- $|B_{z_i}^i| \leq \frac{\alpha-1}{\alpha-2}|S_i| - \frac{\alpha}{\alpha-2}k$,
- $\deg_{T_i}(z_i) \leq \frac{1}{(\alpha-2)k}|S_i| - \frac{2}{\alpha-2}$.

Let T be the tree obtained from the disjoint union of T_1 and T_2 by merging z_1 and z_2 into a vertex z . Let $B_z := B_{z_1}^1 \cup B_{z_2}^2$. Let $B_x := B_x^i$ for each $x \in V(T_i) \setminus \{z_i\}$. Since $G = G_1 \cup G_2$ and $V(G_1 \cap G_2) = B_{z_1}^1 \cap B_{z_2}^2 \subseteq B_z$, we have that $(B_x : x \in V(T))$ is a tree-partition of G . By construction, $S \subseteq B_z$ and since $V(G_1 \cap G_2) \subseteq B_{z_i}^i$ for each i ,

$$\begin{aligned}|B_z| &\leq |B_{z_1}^1| + |B_{z_2}^2| - |V(G_1 \cap G_2)| \\ &\leq \left(\frac{\alpha-1}{\alpha-2}|S_1| - \frac{\alpha}{\alpha-2}k\right) + \left(\frac{\alpha-1}{\alpha-2}|S_2| - \frac{\alpha}{\alpha-2}k\right) - |V(G_1 \cap G_2)| \\ &= \frac{\alpha-1}{\alpha-2}(|S_1| + |S_2|) - \frac{2\alpha}{\alpha-2}k - |V(G_1 \cap G_2)| \\ &\leq \frac{\alpha-1}{\alpha-2}(|S| + 2|V(G_1 \cap G_2)|) - \frac{2\alpha}{\alpha-2}k - |V(G_1 \cap G_2)| \\ &= \frac{\alpha-1}{\alpha-2}|S| - \frac{2\alpha}{\alpha-2}k + \frac{\alpha}{\alpha-2}|V(G_1 \cap G_2)| \\ &\leq \frac{\alpha-1}{\alpha-2}|S| - \frac{2\alpha}{\alpha-2}k + \frac{\alpha}{\alpha-2}k \\ &= \frac{\alpha-1}{\alpha-2}|S| - \frac{\alpha}{\alpha-2}k \\ &\leq \frac{3\alpha(\alpha-1)}{\alpha-2}kd - \frac{\alpha}{\alpha-2}k.\end{aligned}$$

Every other part has the same size as in the tree-partition of G_1 or G_2 . So this tree-partition of G has width at most $\frac{3\alpha(\alpha-1)}{\alpha-2}kd - \frac{\alpha}{\alpha-2}k$. Note that

$$\deg_T(z) = \deg_{T_1}(z_1) + \deg_{T_2}(z_2)$$

$$\begin{aligned}
&\leq \frac{1}{(\alpha-2)k} |S_1| - \frac{2}{\alpha-2} + \frac{1}{(\alpha-2)k} |S_2| - \frac{2}{\alpha-2} \\
&\leq \frac{1}{(\alpha-2)k} (|S_1| + |S_2|) - \frac{4}{\alpha-2} \\
&\leq \frac{1}{(\alpha-2)k} (|S| + 2k) - \frac{4}{\alpha-2} \\
&\leq \frac{1}{(\alpha-2)k} |S| - \frac{2}{\alpha-2} \\
&\leq \frac{3\alpha}{\alpha-2} d - \frac{2}{\alpha-2} \\
&< \frac{3\alpha}{\alpha-2} d + \frac{\alpha-4}{\alpha-2}.
\end{aligned}$$

Every other node of T has the same degree as in T_1 or T_2 . Thus $\Delta(T) \leq \frac{3\alpha}{\alpha-2} d + \frac{\alpha-4}{\alpha-2}$. This completes the proof. $\square$

Lemma 5 with $\alpha = 4$ implies the following slight strengthening of Theorem 2.

Theorem 6. *Every non-trivial graph G has a T -partition of width at most*

$$2(\text{tw}(G) + 1)(9\Delta(G) - 1),$$

for some tree T with $\Delta(T) \leq 6\Delta(G)$.

Lemma 5 with $\alpha = 2 + \sqrt{2}$ (chosen to minimise $\frac{3\alpha(\alpha-1)}{\alpha-2}$) implies the next result.

Theorem 7. *Every non-trivial graph G has a T -partition of width at*

$$(1 + \sqrt{2})(\text{tw}(G) + 1)(3(1 + \sqrt{2})\Delta(G) - 1),$$

for some tree T with $\Delta(T) \leq (3 + 3\sqrt{2})\Delta(G) - 3(\sqrt{2} - 1)$.

Acknowledgements Research of Distel is supported by an Australian Government Research Training Program Scholarship. Research of Wood is supported by the Australian Research Council.

References

- [1] NOGA ALON, GUOLI DING, BOGDAN OPOROWSKI, AND DIRK VERTIGAN. [Partitioning into graphs with only small components](#). *J. Combin. Theory Ser. B*, 87(2):231–243, 2003.
- [2] JÁNOS BARÁT AND DAVID R. WOOD. [Notes on nonrepetitive graph colouring](#). *Electron. J. Combin.*, 15:R99, 2008.
- [3] HANS L. BODLAENDER. [The complexity of finding uniform emulations on fixed graphs](#). *Inform. Process. Lett.*, 29(3):137–141, 1988.
- [4] HANS L. BODLAENDER. [The complexity of finding uniform emulations on paths and ring networks](#). *Inform. and Comput.*, 86(1):87–106, 1990.
- [5] HANS L. BODLAENDER. [A partial \$k\$ -arboretum of graphs with bounded treewidth](#). *Theoret. Comput. Sci.*, 209(1-2):1–45, 1998.

- [6] HANS L. BODLAENDER. [A note on domino treewidth](#). *Discrete Math. Theor. Comput. Sci.*, 3(4):141–150, 1999.
- [7] HANS L. BODLAENDER AND JOOST ENGELFRIET. [Domino treewidth](#). *J. Algorithms*, 24(1):94–123, 1997.
- [8] HANS L. BODLAENDER, CARLA GROENLAND, AND HUGO JACOB. [On the parameterized complexity of computing tree-partitions](#). 2022, arXiv:2206.11832.
- [9] HANS L. BODLAENDER AND JAN VAN LEEUWEN. [Simulation of large networks on smaller networks](#). *Inform. and Control*, 71(3):143–180, 1986.
- [10] RUTGER CAMPBELL, KATIE CLINCH, MARC DISTEL, J. PASCAL GOLLIN, KEVIN HENDREY, ROBERT HICKINGBOTHAM, TONY HUYNH, FRED-DIE ILLINGWORTH, YOURI TAMITEGAMA, JANE TAN, AND DAVID R. WOOD. [Product structure of graph classes with bounded treewidth](#). 2022, arXiv:2206.02395.
- [11] RUTGER CAMPBELL, MARC DISTEL, J. PASCAL GOLLIN, DANIEL J. HARVEY, KEVIN HENDREY, ROBERT HICKINGBOTHAM, BOJAN MOHAR, AND DAVID R. WOOD. [Graphs of linear growth have bounded treewidth](#). 2022, arXiv:tba.
- [12] PAZ CARMİ, VIDA DUJMOVIĆ, PAT MORIN, AND DAVID R. WOOD. [Distinct distances in graph drawings](#). *Electron. J. Combin.*, 15:R107, 2008.
- [13] DIMITRIS CHATZIDIMITRIOU, JEAN-FLORENT RAYMOND, IGNASI SAU, AND DIMITRIOS M. THILIKOS. [An \$O\(\log \text{OPT}\)\$ -approximation for covering and packing minor models of \$\theta_r\$](#) . *Algorithmica*, 80(4):1330–1356, 2018.
- [14] EMILIO DI GIACOMO, GIUSEPPE LIOTTA, AND HENK MEIJER. [Computing straight-line 3D grid drawings of graphs in linear volume](#). *Comput. Geom. Theory Appl.*, 32(1):26–58, 2005.
- [15] GUOLI DING AND BOGDAN OPOROWSKI. [Some results on tree decomposition of graphs](#). *J. Graph Theory*, 20(4):481–499, 1995.
- [16] GUOLI DING AND BOGDAN OPOROWSKI. [On tree-partitions of graphs](#). *Discrete Math.*, 149(1–3):45–58, 1996.
- [17] VIDA DUJMOVIĆ, LOUIS ESPERET, PAT MORIN, BARTOSZ WALCZAK, AND DAVID R. WOOD. [Clustered 3-colouring graphs of bounded degree](#). *Combin. Probab. Comput.*, 31(1):123–135, 2022.
- [18] VIDA DUJMOVIĆ, GWENAËL JORET, PIOTR MICEK, PAT MORIN, AND DAVID R. WOOD. [Bad news for product structure of bounded-degree graphs](#). 2022, arXiv:tba.
- [19] VIDA DUJMOVIĆ, PAT MORIN, AND DAVID R. WOOD. [Layout of graphs with bounded tree-width](#). *SIAM J. Comput.*, 34(3):553–579, 2005.
- [20] VIDA DUJMOVIĆ, MATTHEW SUDERMAN, AND DAVID R. WOOD. [Graph drawings with few slopes](#). *Comput. Geom. Theory Appl.*, 38:181–193, 2007.
- [21] ANDERS EDENBRANDT. [Quotient tree partitioning of undirected graphs](#). *BIT*, 26(2):148–155, 1986.
- [22] JOHN P. FISHBURN AND RAPHAEL A. FINKEL. [Quotient networks](#). *IEEE Trans. Comput.*, C-31(4):288–295, 1982.

- [23] ARCHONTIA C. GIANNOPOULOU, O-JOUNG KWON, JEAN-FLORENT RAYMOND, AND DIMITRIOS M. THILIKOS. [Packing and covering immersion models of planar subcubic graphs](#). In PINAR HEGGERNES, ed., *Proc. 42nd Int'l Workshop on Graph-Theoretic Concepts in Computer Science (WG 2016)*, vol. 9941 of *Lecture Notes in Comput. Sci.*, pp. 74–84. 2016.
- [24] RUDOLF HALIN. [Tree-partitions of infinite graphs](#). *Discrete Math.*, 97:203–217, 1991.
- [25] DANIEL J. HARVEY AND DAVID R. WOOD. [Parameters tied to treewidth](#). *J. Graph Theory*, 84(4):364–385, 2017.
- [26] DIETRICH KUSKE AND MARKUS LOHREY. [Logical aspects of Cayley-graphs: the group case](#). *Ann. Pure Appl. Logic*, 131(1–3):263–286, 2005.
- [27] CHUN-HUNG LIU AND SANG-IL OUM. [Partitioning \$H\$ -minor free graphs into three subgraphs with no large components](#). *J. Combin. Theory Ser. B*, 128:114–133, 2018.
- [28] JEAN-FLORENT RAYMOND AND DIMITRIOS M. THILIKOS. [Recent techniques and results on the Erdős-Pósa property](#). *Discrete Appl. Math.*, 231:25–43, 2017.
- [29] BRUCE A. REED. [Algorithmic aspects of tree width](#). In *Recent advances in algorithms and combinatorics*, vol. 11, pp. 85–107. Springer, 2003.
- [30] NEIL ROBERTSON AND PAUL SEYMOUR. [Graph minors. II. Algorithmic aspects of tree-width](#). *J. Algorithms*, 7(3):309–322, 1986.
- [31] DETLEF SEESE. [Tree-partite graphs and the complexity of algorithms](#). In LOTHAR BUDACH, ed., *Proc. Int'l Conf. on Fundamentals of Computation Theory*, vol. 199 of *Lecture Notes in Comput. Sci.*, pp. 412–421. Springer, 1985.
- [32] DAVID R. WOOD. [Vertex partitions of chordal graphs](#). *J. Graph Theory*, 53(2):167–172, 2006.
- [33] DAVID R. WOOD. [On tree-partition-width](#). *European J. Combin.*, 30(5):1245–1253, 2009.
- [34] DAVID R. WOOD AND JAN ARNE TELLE. [Planar decompositions and the crossing number of graphs with an excluded minor](#). *New York J. Math.*, 13:117–146, 2007.
- [35] RUI-RAY ZHANG AND MASSIH-REZA AMINI. [Generalization bounds for learning under graph-dependence: A survey](#). 2022, arXiv:2203.13534.